STATEMENT OF PURPOSE

Applicant: Anthony Oluwaseun Matthew | **Academic Background:** B.Tech in Mathematics (Operations Research & Computational Mathematics), OAUSTECH

Target Program:	Master of Science (M.Sc.) in Computer Science / Software Engineering
Target Institutions:	[Target Institution, e.g., Politecnico di Milano / University of Bologna / University of Passau]
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1. INTRODUCTION & ACADEMIC FOUNDATION

My journey into advanced computing began with a foundational conviction: the most reliable, fault-tolerant software systems are rooted in mathematical principles. Having completed my Bachelor of Technology in Mathematics at Olusegun Agagu University of Science and Technology (OAUSTECH), specializing in Operations Research and Computational Mathematics, I approach software engineering not merely as the mechanics of writing syntax, but as the rigorous design of deterministic state machines, optimal graph algorithms, and constrained optimization models.

Over the past five years, I have actively bridged this theoretical foundation with real-world production engineering—architecting cross-platform mobile applications, engineering zero-loss financial transaction gateways, and managing decoupled cloud infrastructures. I am now applying to the Master of Science in Computer Science / Software Engineering to deepen my mastery of advanced distributed systems, concurrent software architectures, and automated verification paradigms.

2. BRIDGING COMPUTATIONAL MATHEMATICS & PRODUCTION ENGINEERING

During my undergraduate studies, courses in Numerical Analysis, Optimization Theory, and Discrete Mathematics revealed how computational complexity governs real-world computing. I applied this directly to my software engineering practice. Rather than building systems intuitively, I utilized mathematical principles to eliminate bottlenecks in production platforms.

For example, when architecting the transactional pipeline for an e-commerce platform processing thousands of live payments via the Paystack gateway, I recognized that concurrency and network latency created high risks for duplicate ledger credits and state drift. Applying database transaction isolation theory, I implemented atomic row-level locks (SELECT FOR UPDATE) and cryptographic HMAC-SHA512 webhook signature verification. To ensure zero data discrepancy, I designed an idempotent event-driven queueing architecture with exponential backoff workers. This rigorous mathematical approach eliminated 100% of transaction race conditions and guaranteed ledger consistency across volatile network conditions.

3. PROFESSIONAL EXPERIENCE & LEADERSHIP

My professional progression has reinforced my capacity to engineer systems across the entire decoupled stack. Under my software firm, EmpireCodes, I developed and published a tri-tier mobile and web fleet. Utilizing React Native and Expo SDK, I engineered production mobile applications incorporating offline-first state reconciliation with asynchronous storage and synchronized audio-haptic micro-interactions. On the backend, I designed high-throughput RESTful APIs using Python and Django REST Framework, containerized with Docker and deployed across cloud

microservices.

In addition to building commercial products, I served as a Backend Developer and Instructor at AnchorSoft Academy, where I mentored aspiring software engineers in relational database normalization, API security, and Python development. Furthermore, during my National Youth Service Corps (NYSC) deployment within the Procurement Unit of the Somolu Local Government Council (completed 2026), I identified severe manual record-keeping bottlenecks and engineered an automated municipal budget tracking pipeline using Python and relational database schemas, reducing procurement processing turnaround by 40%.

4. WHY THIS MASTER'S PROGRAM & CAREER OBJECTIVES

While my industry experience has given me deep practical fluency with modern developer tools and frameworks, I recognize that solving next-generation engineering problems—such as large-scale distributed consensus, autonomous agent orchestration, and formal verification—demands advanced graduate research.

Your Master of Science curriculum stands out as the ideal intellectual environment for my aspirations. I am particularly drawn to your department's curriculum in Distributed Systems, Concurrent Software Architecture, and Database Systems Optimization. The program's balance of rigorous theoretical foundations and applied engineering aligns directly with my goal of becoming a Lead Systems Architect capable of engineering resilient, high-volume infrastructure.

5. CONCLUSION

I bring a unique blend of theoretical mathematical training, five years of verified production engineering experience, and a relentless drive for excellence. I am fully prepared to meet the academic and collaborative demands of your graduate faculty, and I look forward to contributing actively to your research community.

Thank you for considering my application.

Respectfully submitted,

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